

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (2 hours) April, 2nd, 2016.

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. Dawn-Holly plants have male and female genders. The time for a single female Dawn-Holly plant grown alone to bloom from when it is born (i.e. emergence from the ground) in weeks is distributed exponential with $\lambda=10$; $f(t) = (1/10) e^{-t/10}$. The time for a single male Dawn-Holly grown alone to bloom is exponential with parameter $\lambda=15$; $(1/15) e^{-t/15}$. However, if a female plant is grown paired with (i.e. within 50 feet of) a male plant the time to blooming for both plants is independent except that once the male plant blooms, the pollen released by the male triggers the female plant to immediately bloom if the female plant did not bloom first. Thus when a male and female Dolly-Holly are paired together, the time for the female plant to bloom is $\min(Y, Z)$ where $Y \sim \text{exponential}(\lambda=10)$ and $Z \sim \text{exponential}(\lambda=15)$.

- A. Compared to growing a female Dolly-Holly alone, by how much time on average does pairing the female plant with a male Dolly-Holly plant reduce the expected time to bloom for a female Dolly-Holly?
 - B. If $N=100$ independent pairs of female and male Dolly-Holly plants are grown, Let X be then number of pairs where the male Dolly-Holly bloomed first and stimulated the female plant to bloom. What is the probability that $X \geq 50$?
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2. Liquor stores and convenience stores are frequently targeted by robbers. A criminologist wants to compare the effectiveness of three commonly used alternative methods (M1, M2, M3.) of police car patrolling of these stores. A large city police department agrees to cooperate in the conduct of the experiment. The city has multiple police "precinct" stations each serving a section of the city. 12 precincts are randomly picked to participate in the study, and the criminologist randomly assigns 4 precincts to Method M1, 4 to method M2, and 4 to M3. For each precinct in the study, 2 liquor stores and 2 convenience stores are randomly picked. The study continues for one month, and for each individual store, Y_{ijk} the number of robberies is recorded.

- a) Show the data layout for this study.
 - b) Show an ANOVA table listing Source of variability, degrees of freedom, Expected Mean Squares, and which F ratios you would use to test the Sources of Variability.
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3. A therapist wants to compare two methods for reducing stress. One method is Yoga, and the other is Reiki therapy. For 24 suitable patients, the therapist measures their initial stress level, and randomly refers 12 to a Yoga therapist, and the other 12 to a Reiki therapist. After 3 months of therapy the stress levels of the 24 patients are again measured. Stress is measured on a psychological scale that is standardized to a normal distribution. A summary of The Stress Results is given on the next page:

	[FOR REIKI GROUP, n=12]			[FOR YOGA GROUP, n=12]		
	Pre-Score	Post-Score	Difference	Pre-Score	Post-Score	Difference
Mean	75.7	26.3	49.3	79.1	34.8	44.3
Std-Dev	9.7	3.8	8.4	8.6	9.0	2.9

a) The researcher realizes that there is a difference in the Pre stress values, and wants to adjust for this. Carry out a t-test to compare the Effect of the Therapy Methods that adjusts for treatment group differences in the pre values. Show the formula you used. Give a corresponding confidence interval for the difference in the average effects of Reiki vs. Yoga.

b) A different approach to adjust for the Pre-stress level is to use Post-stress score as the response variable, and the Pre-stress score as a covariate. A statistician fit the following least-squares model:

Where therapy = -1 for Yoga; therapy = 1 for Reiki.

Post-stress Score is the Response, Pre-stress Score is in the model as a predictor covariate

Term	Estimate	Stand error	t	Prob> t	
Intercept	-16.610804	5.8913993	-2.8195007	0.01058995	
Therapy	-3.18221	0.67027342	-4.7476297	0.00012306	
Pre	0.60065756	0.07543852	7.96221307	0.00000013	
(pre-77.375)*Therapy	-0.396054	0.07543852	-5.2500233	0.00003879	
	Df	SS	MS	F	Prob> F
Model	3	1274.28253	424.760844	40.9061474	<0.001
Error	20	207.6758	10.38379		
C. Total	23	1481.95833	.	.	

Interpret the information in part (b) in a way that a non-statistical researcher can understand. Which is more effective in reducing stress, Reiki or Yoga; in what ways and under what circumstances and for what types of patients?

c) Explain to the researcher how to reconcile the conclusions from part (a) and part (b). Is there a difference in the stress reducing effect of Reiki and Yoga?

4. You read a journal article on TOTAL-cholesterol, HDL-cholesterol and LDL-cholesterol from a study of **n=100 patients** where TOTAL-cholesterol = HDL + LDL cholesterol. Assume that for HDL, LDL and TOTAL cholesterol, the distributions are iid multivariate normal. The article reports for:

HDL: sample-mean = 145 and p-value for the population mean to be 150 is 0.10

LDL: sample-mean = 48 and p-value for the population mean to be 50 is 0.05

TOTAL: sample-mean = 193 and p-value for the population mean to be 200 is 0.05

All P-values are Two-Sided

You must derive the standard deviations and covariances for HDL-cholesterol and LDL-cholesterol in a single observation from this information

- A. Work backwards from the p-values for the population means be equal to the specified values based on the observed sample means to obtain the standard deviations for both HDL-cholesterol and LDL-cholesterol in a single observation?
- B. What is the covariance of HDL and LDL cholesterol in a single observation?

